EC4215 · BUSINESS ECONOMETRICS I

Full Marking Scheme

Winter 2024 Examination · Paper 1

Institution	University College Cork (NUI, Cork)
Internal Examiner	Dr Brian Turner
External Examiner	Dr Kevin Mulligan
Head of School / Department	Prof Justin Doran
Session	Winter 2024
Duration	1.5 hours
Structure	Section A: Long Questions — answer **2 out of 3** (100% of total marks; all questions carry equal weight)
Materials Permitted	Non-programmable calculators may be used

Coverage. This scheme answers all three long questions (A1, A2, A3) in full. Under exam conditions you choose your best two, so full answers to all three let you prepare every route through the paper and pick strategically on the day.

Source basis. The exam paper text, regression outputs (Tables 3.1–3.3), and all statistical tables (Appendices A1–A5) are taken verbatim from the uploaded Winter 2024 paper. All t-statistics, degrees of freedom, critical values and the Durbin–Watson comparison below were computed and cross-checked independently against the paper's own appendix tables before being written here. Where an answer draws on standard econometric theory beyond the paper itself (e.g. the four criteria for including a variable), this follows the standard treatment in introductory econometrics texts (Studenmund; Gujarati) and is flagged inline where relevant.

Question A1 Regression interpretation · research design · sampling (100%)

Part (a) — Gondola rides regression (60%)

Question (a): A researcher examines the weekly demand for gondola rides in Venice over a year and produces (standard errors in brackets):

$$G_t = 7.8 + 1.4T_t - 1.2R_t + 8.4S_t \quad (0.4)\ (0.3)\ (5.8) \quad N = 52, \text{Adjusted } R^2 = 0.71$$

where G = gondola rides (000s), T = tourists (000s), R = rainfall (mm), S = summer dummy (June–August).
(i) Carefully interpret the results from this regression. **(35%)**
(ii) Does the equation fit your expectations? Why or why not? **(15%)**
(iii) If you could add one more variable, what would it be and why? **(10%)**

FULL MODEL ANSWER — A1(A)(I)

Coefficient interpretation (units matter — both G and T are in thousands). The intercept (7.8) says that in a week with zero tourists, zero rainfall and outside summer, predicted gondola rides are 7,800 — a mechanical baseline with little standalone economic meaning since T = 0 is far outside the sample. The tourist coefficient (1.4) says that, holding rainfall and season constant, each additional 1,000 tourists in a week is associated with 1,400 additional gondola rides. The rainfall coefficient (−1.2) says each additional millimetre of weekly rainfall is associated with 1,200 fewer rides, ceteris paribus. The summer dummy (8.4) says a summer week (June–August) has 8,400 more rides than an otherwise identical non-summer week — an intercept shift, not a slope change.

Statistical significance (compute t = coefficient / standard error). With N = 52 and k = 4 estimated parameters, df = 48; the 5% two-tailed critical value from the paper's own t-table is approximately 2.01 (between df = 40 and df = 50 rows).

Variable	Coefficient	Std. error	t-statistic	vs $\lvert t^*\rvert \approx 2.01$
T (tourists)	1.4	0.4	3.50	Significant at 5%
R (rainfall)	−1.2	0.3	−4.00	Significant at 5%
S (summer dummy)	8.4	5.8	1.45	Not significant

Overall fit. Adjusted R^2 = 0.71: about 71% of the variation in weekly gondola rides is explained by the model after adjusting for degrees of freedom — a strong fit for weekly demand data. N = 52 is one full year of weekly observations, so the sample covers exactly one seasonal cycle.

The subtle point examiners reward: the summer dummy is individually insignificant even though its coefficient is large. A likely reason is that summer weeks are also high-tourist weeks — T and S move together, inflating the standard error on S. Noting this (rather than just mechanically labelling S "insignificant") is what separates a first-class answer.

Mark Allocation — Part (a)(i) (35%)

Band	Criteria
Full credit (30–35)	Correctly interprets all four coefficients with correct units (thousands for G and T) AND computes all three t-statistics (3.50, −4.00, 1.45) with correct df (48) and critical value (≈2.01) AND interprets adjusted R^2 AND correctly interprets the dummy as an intercept shift.
Good (24–29)	Interprets all coefficients and computes t-statistics correctly, but slips on units, df, or gives a thin R^2 interpretation.
Partial (14–23)	Interprets signs/magnitudes sensibly but omits significance testing, or tests significance without interpreting coefficients in units.
Minimal (0–13)	Restates the equation, misreads the dummy as a slope, or interprets coefficients without ceteris paribus logic.

FULL MODEL ANSWER — A1(A)(II)

Yes — signs match prior expectations, with one nuance. Tourists (+): gondola rides are a tourist activity, so demand should rise with tourist numbers — confirmed and strongly significant. Rainfall (−): gondola rides are an outdoor, discretionary activity; rain deters them — confirmed and significant. Summer (+): we would expect summer to boost rides (peak season, better weather, longer days) — the sign is positive as expected, but it is insignificant once tourists and rainfall are controlled for. That is itself economically sensible: much of what "summer" means for gondola demand is *already captured* by higher tourist numbers and lower rainfall, so the dummy has little independent work to do. The strong adjusted R^2 (0.71) is also plausible for demand driven largely by observable footfall and weather.

Mark Allocation — Part (a)(ii) (15%)

Band	Criteria
Full credit (13–15)	States expected sign for each variable with a one-line economic justification AND reconciles the insignificant summer dummy (its effect is largely absorbed by T and R).
Good (10–12)	Correct expectations and justifications for all three variables but no reconciliation of the insignificant dummy.
Partial (6–9)	Expectations stated for some variables with weak or missing justification.
Minimal (0–5)	Yes/no assertion with no economic reasoning.

FULL MODEL ANSWER — A1(A)(III)

One well-justified variable is all that's needed. Strongest candidate: the price of a gondola ride (average weekly price, €). Demand theory says own price is a first-order determinant of quantity demanded, and omitting it risks omitted-variable bias if price correlates with included regressors (prices likely rise in high-tourist summer weeks — meaning the current tourist/summer coefficients may partly pick up price effects). Expected sign: negative.

Other creditable choices (any one, well justified): temperature (weather affects outdoor leisure beyond rainfall alone); price of substitute activities (vaporetto/water-taxi fares); cruise-ship arrivals (a distinct, lumpy source of day-trippers); a holiday/festival dummy (Carnevale, Regata Storica). The mark is for the *justification* — expected sign, economic mechanism, and ideally the omitted-variable-bias logic — not the specific variable named.

Mark Allocation — Part (a)(iii) (10%)

Band	Criteria
Full credit (9–10)	Names a sensible variable AND gives expected sign AND explains the economic mechanism, ideally noting the omitted-variable-bias rationale.
Good (7–8)	Sensible variable with expected sign and brief mechanism.
Partial (4–6)	Sensible variable, weak or missing justification.
Minimal (0–3)	Implausible variable or no justification.

Part (b) — Study time and grades: research design (40%)

Question (b): The President of UCC asks you to research whether time spent studying affects grades. You talk to a dozen of your classmates, find a positive correlation between study time and grades, report back, and go for a cup of coffee.
(i) Would this represent best practice for this type of research? Explain. **(15%)**
(ii) Having had the coffee, how would you re-do the exercise? Explain. **(25%)**

FULL MODEL ANSWER — A1(B)(I)

No — this fails best practice on essentially every dimension. (1) **Tiny sample:** N = 12 gives almost no statistical power; sampling error dominates and no reliable inference is possible. (2) **Non-random, unrepresentative sample:** your own classmates are a convenience sample — same course, same year, same institution, and plausibly similar in motivation and ability to you. Results cannot generalise to UCC students at large, which is the population the President cares about. (3) **Correlation is not causation:** a raw bivariate correlation cannot establish that studying *affects* grades. Confounders (ability, motivation, attendance, prior attainment) plausibly drive both study time and grades, and reverse causality is possible (students doing badly may study more, or high achievers may enjoy studying). (4) **Measurement quality:** casual self-reports of study time from a chat are noisy and prone to reporting bias. (5) **No controls, no model:** nothing else that affects grades is held constant.

Mark Allocation — Part (b)(i) (15%)

Band	Criteria
Full credit (13–15)	Says no AND raises at least three distinct flaws including (sample size or representativeness) AND correlation-vs-causation with a named confounder or reverse-causality story.
Good (10–12)	Says no with two well-explained flaws.
Partial (6–9)	Identifies one flaw clearly, or lists flaws without explanation.
Minimal (0–5)	Yes, or no with no substantive reasoning.

FULL MODEL ANSWER — A1(B)(II)

Design a proper study — the answer should hit sampling, measurement, and modelling.

1. Define the population and sample properly. Population: all UCC students (or a clearly defined subset). Draw a large *random* sample — e.g. several hundred students sampled across faculties, years and programmes (stratified random sampling would ensure representation of each faculty/year). Larger N raises power and shrinks standard errors.

2. Measure carefully. Collect study time via a structured survey or, better, time-use diaries over several weeks rather than one casual question; obtain actual grades (with consent, from records rather than self-reports); collect controls: prior attainment (e.g. CAO points), attendance, hours of paid work, programme, year of study.

3. Model rather than correlate. Estimate a multiple regression, e.g.

$$\mathrm{GRADE}_i = \beta_0 + \beta_1\mathrm{STUDY}_i + \beta_2\mathrm{ABILITY}_i + \beta_3\mathrm{ATTEND}_i + \beta_4\mathrm{WORK}_i + \ldots + \varepsilon_i$$

so the study-time effect is estimated *holding constant* the main confounders, then test H_0: $\beta_1 = 0$ with a t-test. Check robustness (functional form — study time may have diminishing returns, so consider a quadratic; outliers; heteroskedasticity).

4. Be honest about causality. Even the improved observational design identifies association conditional on controls, not watertight causation. A first-class answer notes what would strengthen causal claims: panel data following the same students over time, or an experiment/natural experiment (e.g. random assignment to a study-skills programme as an instrument for study time).

Mark Allocation — Part (b)(ii) (25%)

Band	Criteria
Full credit (21–25)	Large random/representative sample AND improved measurement of both study time and grades AND a multiple regression with named controls and a hypothesis test AND acknowledges remaining causality limits (panel/experimental ideas earn the top of the band).
Good (16–20)	Random larger sample plus regression with controls; measurement or causality discussion thin.
Partial (9–15)	Improves the sample (bigger/more representative) but stays with correlation, or proposes regression without sampling improvements.
Minimal (0–8)	Superficial changes (e.g. "ask more classmates") with no design logic.

Question A1 total **35 + 15 + 10 + 15 + 25 = 100%**

Question A2 Variable inclusion decision · specification issues in time series (100%)

Part (a) — Should PHB be included? (70%)

Question (a): A researcher examines take-up of private health insurance (standard errors in brackets):

Eqn 2.1: $PHI_t = 257.1 - 12.7PR_t + 24.7WT_t + 157.8INC_t$ (4.3) (10.1) (43.4) N = 25, Adj. R^2 = 0.647

Eqn 2.2: $PHI_t = 205.1 - 13.5PR_t + 23.4WT_t + 156.7INC_t + 12.7PHB_t$ (4.3) (9.8) (42.1) (3.6) N = 25, Adj. R^2 = 0.662

where PHI = people with private health insurance (000s), PR = health-insurance CPI, WT = people waiting >6 months for public inpatient treatment (000s), INC = average annual income (€000), PHB = private hospital beds.
Should the researcher include PHB_t (i.e. use Eqn 2.2 instead of Eqn 2.1)? Explain. **(70%)**

FULL MODEL ANSWER — A2(A)

Frame the answer around the standard criteria for deciding whether a variable belongs in a regression: (1) theory — is there a sound economic rationale? (2) t-test — is the new variable individually significant? (3) adjusted R^2 — does overall explanatory power improve? (4) do the other coefficients change materially when it is added (a sign the new variable was correcting omitted-variable bias)? Work through each in turn.

1. Theory. Private hospital bed capacity is a plausible driver of insurance take-up: more private capacity means insurance buys real access to treatment, raising its value to households (a supply-side complement to insurance demand). Causality could partly run the other way (insurers expand capacity where membership grows), which is worth acknowledging, but a positive theoretical link exists. Expected sign: positive — and the estimated coefficient (+12.7) matches.

2. Significance of the new variable. t = 12.7 / 3.6 = 3.53. With N = 25 and k = 5 parameters in Eqn 2.2, df = 20, and the 5% two-tailed critical value from the paper's t-table is 2.086. Since 3.53 > 2.086, PHB is significant at the 5% level (indeed also at 1%, where t* = 2.845).

3. Adjusted R^2. Rises from 0.647 to 0.662. Because *adjusted* R^2 penalises extra regressors, an increase means PHB adds explanatory power beyond its degrees-of-freedom cost. (Plain R^2 would rise mechanically; the adjusted rise is the informative one.)

4. Stability of the other coefficients.

Variable	Eqn 2.1 coeff. (t)	Eqn 2.2 coeff. (t)	Change
PR (price index)	−12.7 (−2.95)	−13.5 (−3.14)	Small; stays negative & significant
WT (waiting lists)	24.7 (2.45)	23.4 (2.39)	Small; stays positive & significant
INC (income)	157.8 (3.64)	156.7 (3.72)	Essentially unchanged
PHB (private beds)	—	12.7 (3.53)	**Significant at 5% (and 1%)**

Slope coefficients barely move and all keep their signs and significance (Eqn 2.1: df = 21, t* = 2.080; Eqn 2.2: df = 20, t* = 2.086 — every reported slope clears its critical value in both equations). The intercept falls from 257.1 to 205.1, which is unproblematic. Coefficient stability means adding PHB does not destabilise the model — and standard errors on WT and INC actually *fall* slightly, so no multicollinearity problem has been introduced.

Verdict: Yes — include PHB and use Eqn 2.2. All four criteria point the same way: sound theory with the expected sign, individually significant (3.53 vs 2.086), adjusted R^2 up, and stable coefficients elsewhere. Conversely, leaving out a relevant variable risks omitted-variable bias in the retained coefficients, so exclusion is the riskier choice. A top answer may add the caveats that N = 25 is small, and that with time-series data one would also want to check residual autocorrelation before fully signing off.

Check note. The four-criteria framework (theory, t-test, adjusted R^2, coefficient stability) is the standard textbook decision rule for adding a variable (Studenmund's "four important specification criteria"). The paper does not print the framework itself; every number in the table above is either printed on the paper or computed as coefficient ÷ standard error and verified against the paper's own Appendix A3 critical values.

Mark Allocation — Part (a) (70%)

Band	Criteria
Full credit (60–70)	Organises the answer around explicit inclusion criteria AND computes t for PHB (3.53) with correct df (20) and critical value (2.086) AND interprets the adjusted-R^2 rise correctly (noting the penalty logic) AND examines coefficient stability across the two equations AND gives a clear, justified verdict with the theoretical rationale for PHB.
Good (46–59)	Correct verdict using at least three criteria with correct t-test; stability or theory treated thinly.
Partial (28–45)	Correct verdict resting on one or two criteria (e.g. t-test alone, or "R^2 went up") with limited reasoning, or minor computational slips.
Minimal (0–27)	Verdict without evidence, misreads adjusted R^2, or compares raw coefficient sizes as the test.

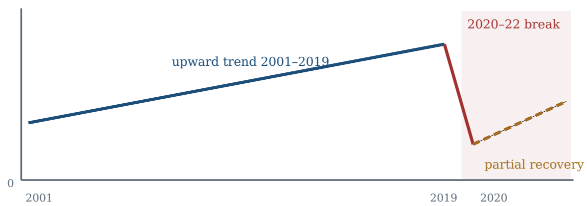
Part (b) — Time-series scatter: specification issue (30%)

Question (b): You are given an annual time series for a dependent variable, 2001–2022 (Figure 2.1): values climb from ~14 in 2001 to ~26 by 2018–19, then collapse to ~11 in 2020 before partially recovering by 2022.
(i) What potential specification issue is exemplified by this graph, and what are the consequences? **(15%)**
(ii) If the dependent variable is the number of restaurant bookings in Ireland (millions), how would you deal with the issue? **(15%)**

FULL MODEL ANSWER — A2(B)(I)

The graph shows two related specification problems. First, the series **trends steadily upward** from 2001 to 2019 — the tell-tale sign of an **omitted time trend**: some slow-moving force (population, income, taste change) is pushing the variable up over time. Second, there is an abrupt **structural break** in 2020–21: the level collapses far below trend and only partially recovers — a break in the data-generating process, not ordinary noise.

Consequences if ignored. If the model omits the trend, any trending regressor will soak up the trend's effect — **omitted-variable bias** — and two series that merely trend together will appear strongly related when no genuine relationship exists (**spurious regression**: inflated R^2 and t-statistics, typically with a very low Durbin–Watson statistic). Ignoring the 2020 break means the fitted line averages across two different regimes: coefficient estimates are distorted, the residuals around 2020–22 are huge and systematic (autocorrelated), and forecasts fail.



Stylised view of Figure 2.1: a steady upward trend, then a sharp 2020 collapse below trend with partial recovery — trend plus structural break.

Mark Allocation — Part (b)(i) (15%)

Band	Criteria
Full credit (13–15)	Identifies the omitted time trend AND the 2020 structural break AND gives consequences: omitted-variable/spurious-regression bias, distorted coefficients, autocorrelated residuals, unreliable forecasts.
Good (10–12)	Identifies one issue fully with correct consequences, and gestures at the other.
Partial (6–9)	Identifies an issue but consequences vague ("results will be wrong").
Minimal (0–5)	Describes the picture without naming a specification issue.

FULL MODEL ANSWER — A2(B)(II)

Restaurant bookings in Ireland make the 2020 story concrete: the collapse is COVID-19 — lockdowns closed restaurants in 2020–21 and the sector only partially recovered by 2022. Two concrete fixes, used together:

1. Add a deterministic time trend (t = 1, 2, …, 22) to capture the steady growth in bookings from rising population, incomes and eating-out habits over 2001–2019, so other coefficients aren't contaminated by trend effects.

2. Add a COVID dummy equal to 1 in the affected years (2020 and 2021, possibly a separate or scaled dummy for partial-recovery 2022) and 0 otherwise, capturing the level shift from lockdown restrictions. Both new terms should then be tested for significance in the usual way.

Alternatives worth credit: estimating the model only on the stable pre-2020 sample (defensible but discards information); including a direct restrictions-stringency variable instead of a plain dummy; or interacting the dummy with slopes if the *relationships* (not just the level) changed during COVID. What earns full marks is matching the tool to the diagnosis: trend term for the trend, dummy for the break.

Mark Allocation — Part (b)(ii) (15%)

Band	Criteria
Full credit (13–15)	Recognises COVID as the 2020 cause AND proposes a time trend for the trend AND a correctly-defined dummy (1 in 2020–21, 0 otherwise) for the break, with the logic for each.
Good (10–12)	Proposes both fixes but with loose dummy definition or thin reasoning, or one fix explained excellently.
Partial (6–9)	One sensible fix, weakly justified.
Minimal (0–5)	No workable fix (e.g. "collect better data") or fails to connect 2020 to COVID.

Question A2 total **70 + 15 + 15 = 100%**

Question A3 Multicollinearity: detection, diagnosis and remedies (100%)

Part (a) — Apples demand equation (30%)

Question (a): Demand for apples in a supermarket chain over time:

$$QA_t = \beta_0 + \beta_1 PA_t + \beta_2 PO_t + \beta_3 PP_t + \beta_4 PS_t + \beta_5 INC_t + \varepsilon_t$$

QA = apples purchased (000s); PA, PO, PP, PS = price indices for apples, oranges, pineapples, sweets (2016 = 100); INC = average household income (€000).
(i) Does this equation have possible sources of multicollinearity? Which variables might cause it? Explain. **(15%)**
(ii) Outline how you might address this multicollinearity. **(15%)**

FULL MODEL ANSWER — A3(A)(I)

Yes — this specification is a classic multicollinearity risk, for two reasons. First, it contains **four price indices** (PA, PO, PP, PS) in a time-series setting. All consumer price indices tend to drift upward together with general inflation, so the four price series will be highly correlated with one another over time — they share a common inflationary trend. PO and PP (oranges and pineapples — both fruit, both substitutes for apples) are especially likely to move together as fruit supply-and-cost conditions are shared. Second, **income (INC) also trends upward over time** in nominal terms, correlating with all of the price indices — nominal incomes and prices rise together. So the danger is not one pair of variables but a web of near-linear relationships among five trending regressors. The consequence: individual coefficients become imprecisely estimated (inflated standard errors, low t-stats) even if the overall fit is high, and estimates become unstable across specifications.

Mark Allocation — Part (a)(i) (15%)

Band	Criteria
Full credit (13–15)	Says yes AND identifies the price indices as jointly trending with inflation AND flags INC trending with nominal prices AND explains the mechanism (shared time trends → high pairwise correlation) with the standard-error consequence.
Good (10–12)	Identifies the price-index cluster with sound reasoning; INC link or consequences underdeveloped.
Partial (6–9)	Names some correlated variables without explaining why they correlate.
Minimal (0–5)	Yes/no without identifying variables, or confuses multicollinearity with other problems.

FULL MODEL ANSWER — A3(A)(II)

Standard remedies, matched to this equation. (1) **Do nothing** — legitimate if the goal is forecasting or the key coefficient (own price PA) remains significant; multicollinearity does not bias estimates, it only widens standard errors. (2) **Drop redundant variables** — e.g. keep one substitute price (oranges) and drop pineapples/sweets if theory permits; risk: omitted-variable bias if a dropped variable truly belongs. (3) **Transform variables to break the common trend** — deflate to *relative* prices (e.g. PA/PO or real prices deflated by CPI) and use real income; ratios and deflated series carry the economically meaningful variation without the shared inflation trend. First-differencing is another trend-removing option. (4) **Combine variables** — a single "other fruit price" index averaging PO and PP. (5) **Increase the sample** — more observations add independent variation and shrink standard errors. Best answers note the theory-first principle: don't drop a variable demand theory says must be there (own price) just to lower a VIF.

Mark Allocation — Part (a)(ii) (15%)

Band	Criteria
Full credit (13–15)	At least three distinct remedies, each with rationale and a downside/caveat, including a transformation tailored to this equation (relative/real prices) and the "do nothing" option with its justification.
Good (10–12)	Two or three remedies correctly explained; caveats thin.
Partial (6–9)	One or two remedies listed with little reasoning.
Minimal (0–5)	Vague ("fix the data") or incorrect remedies.

Part (b) — House price regression, Table 3.1 (40%)

Question (b): Quarterly Irish house prices (RESPRICE, Jan 2005 = 100) regressed on new dwelling completions (COMPLET), employment (EMPLOY, 000s), personal consumption (CONSUM, €m), GDP (€m) and CPI, estimated on EViews: sample 2012Q1–2018Q4, N = 28. Key output — $R^2 = 0.9845$, Adj. $R^2 = 0.9810$, F = 279.54 (p = 0.0000), DW = 0.835; t-stats: C 0.72, COMPLET 2.48, EMPLOY 5.65, CONSUM −1.37, GDP 1.53, CPI −1.59.
(i) Discuss the specification of the model. **(15%)**
(ii) Interpret the results. As expected, or anything surprising? **(15%)**
(iii) Do you suspect multicollinearity in this regression? Explain. **(10%)**

FULL MODEL ANSWER — A3(B)(I)

Assess the specification on theory, variable choice, functional form and data. Theory/variable choice: the regressors are sensible demand-and-supply drivers of house prices — employment and consumption/GDP capture demand-side purchasing power and macro conditions, completions capture new supply, CPI captures the general price level (distinguishing real from nominal house-price movements). **Gaps a strong answer flags:** no interest-rate/mortgage-credit variable (a first-order driver of housing demand), no population or household-formation variable, and no housing stock measure; completions measure the *flow* of new supply only. **Redundancy:** including both CONSUM and GDP is questionable — consumption is a large component of GDP, so they overlap heavily (a multicollinearity red flag by construction). **Functional form and data:** a linear levels specification on quarterly macro time series in which nearly every variable trends upward over 2012–2018 (the recovery period) invites common-trend problems; N = 28 is small for five regressors; quarterly data may also need seasonal treatment (completions are seasonal).

Mark Allocation — Part (b)(i) (15%)

Band	Criteria
Full credit (13–15)	Evaluates theoretical relevance of included variables AND names at least one important omission (interest rates/credit, population, stock) AND flags the CONSUM–GDP overlap AND comments on levels/trending data or small N.
Good (10–12)	Sound discussion of included variables plus one further specification point (omission or overlap).
Partial (6–9)	Describes what the variables are with limited evaluation.
Minimal (0–5)	Restates the variable list.

FULL MODEL ANSWER — A3(B)(II)

Individual coefficients (df = 28 − 6 = 22, 5% two-tailed t* = 2.074). Only two regressors are individually significant: EMPLOY (t = 5.65) — positive as expected, more employment means more purchasing power and housing demand — and COMPLET (t = 2.48). CONSUM (−1.37), GDP (1.53) and CPI (−1.59) are all insignificant.

The surprises worth naming. (1) **COMPLET is significantly *positive*:** supply logic says more completions should *depress* prices, so a positive coefficient is the "wrong" sign at face value — best read as completions responding to the same booming demand that drives prices (simultaneity/endogeneity), not as new supply raising prices. (2) **CONSUM and CPI carry negative signs** (though insignificant) where positive links might be expected. (3) **The classic symptom pattern:** $R^2 = 0.9845$ and a hugely significant F (279.5, $p < 0.001$) while three of five regressors are individually insignificant — the model as a whole explains almost everything yet can't attribute it to individual variables. That combination is the textbook fingerprint of severe multicollinearity. (4) **DW = 0.835:** with n = 28 and k = 6 estimated parameters, the paper's Appendix A5 gives $d_L = 1.028$; since $0.835 < d_L$, there is significant positive autocorrelation in the residuals — a further specification warning (consistent with common trends), which undermines the reliability of the reported standard errors and t-tests.

Mark Allocation — Part (b)(ii) (15%)

Band	Criteria
Full credit (13–15)	Tests coefficients against the correct critical value (t* = 2.074, df = 22) AND flags the wrong-signed significant COMPLET with an economic story AND identifies the high-R^2/significant-F/insignificant-t pattern AND notes the low DW ($0.835 < d_L$) as positive autocorrelation.
Good (10–12)	Correct significance testing and at least two of the three "surprises" (COMPLET sign, R^2/t contrast, DW).
Partial (6–9)	Reads off significant/insignificant variables without engaging with signs or the fit/t contrast.
Minimal (0–5)	Describes R^2 only, or misclassifies significance.

FULL MODEL ANSWER — A3(B)(III)

Yes — strongly suspected, on three pieces of indirect evidence already visible in Table 3.1. (1) Very high R^2 (0.9845) and overwhelming joint significance (F = 279.5) combined with mostly insignificant individual t-statistics — collinear regressors share explanatory power, so none looks individually important. (2) The included variables are precisely the kind that move together: EMPLOY, CONSUM and GDP all track the same 2012–2018 economic recovery, and CONSUM is a component of GDP by construction. (3) The implausible/unstable signs (negative CONSUM and CPI) are characteristic of collinear estimation, where coefficients become erratic. Conclusion: suspect severe multicollinearity, to be confirmed with a correlation matrix and VIFs — exactly what part (c) provides.

Mark Allocation — Part (b)(iii) (10%)

Band	Criteria
Full credit (9–10)	Yes, citing the high-R^2-with-insignificant-t pattern AND the economic reason the regressors co-move (shared recovery trend / CONSUM⊂GDP) AND notes formal confirmation requires correlations/VIFs.
Good (7–8)	Yes with the R^2/t-statistic argument correctly explained.
Partial (4–6)	Yes with a partially correct symptom (e.g. only "variables seem related").
Minimal (0–3)	No, or yes without evidence.

Part (c) — Correlation matrix and VIFs, Tables 3.2–3.3 (30%)

Question (c): A correlation matrix (Table 3.2) and centered VIFs (Table 3.3) were generated for the part (b) regression.
(i) Based on these, discuss the presence of multicollinearity. **(15%)**
(ii) What steps would you take to deal with it? Explain. **(15%)**

FULL MODEL ANSWER — A3(C)(I)

Both diagnostics confirm the part (b) suspicion — severe multicollinearity. In the correlation matrix, the pairwise correlations among the "real economy" regressors are extreme: EMPLOY–GDP 0.966, COMPLET–EMPLOY 0.930, COMPLET–GDP 0.912, COMPLET–CONSUM 0.871, CONSUM–GDP 0.839, EMPLOY–CONSUM 0.832 — all far above the common 0.8 warning threshold. Only CPI is moderately correlated with the rest (0.31–0.57).

Variable	Centered VIF	Reading (rule of thumb: VIF > 10 = severe; VIF = $1/(1-R^2_{aux})$)
EMPLOY	23.16	**Severe — ~96% of its variation explained by the other regressors**
GDP	18.35	**Severe**
COMPLET	11.09	**Severe (just above 10)**
CONSUM	5.35	**Moderate**
CPI	2.18	**Low — not part of the problem**

Three of five VIFs exceed 10, with EMPLOY worst at 23.2 (its auxiliary $R^2 \approx 1 - 1/23.2 \approx 0.96$). Verdict: the macro recovery variables (EMPLOY, GDP, COMPLET, and to a lesser degree CONSUM) form one collinear block; CPI stands apart. This explains everything in part (b): the inflated standard errors, insignificant t-stats and unstable signs.

Mark Allocation — Part (c)(i) (15%)

Band	Criteria
Full credit (13–15)	Cites specific high correlations (e.g. EMPLOY–GDP 0.966) against a stated threshold AND applies the VIF > 10 rule identifying EMPLOY (23.2), GDP (18.4), COMPLET (11.1) AND notes CPI is unaffected AND connects the diagnosis back to part (b)'s symptoms.
Good (10–12)	Correctly reads both tables with thresholds, but doesn't link back to (b) or distinguish CPI.
Partial (6–9)	Uses only one diagnostic correctly, or asserts multicollinearity citing numbers loosely.
Minimal (0–5)	Misreads the tables or gives no numeric evidence.

FULL MODEL ANSWER — A3(C)(II)

Remedies, targeted at the diagnosed collinear block. (1) **Drop one or more of the overlapping macro variables** — the strongest candidate is dropping GDP or CONSUM (they overlap by construction; keep whichever theory prioritises, likely leaving EMPLOY as the labour-market demand proxy). Re-estimate and check that VIFs fall and the retained coefficients stabilise. Caveat: dropping a genuinely relevant variable trades multicollinearity for omitted-variable bias — justify the choice on theory. (2) **Transform the data:** estimate in first differences or growth rates, which removes the common 2012–2018 recovery trend that binds the block together (and would likely also improve the DW = 0.835 autocorrelation problem); or deflate nominal aggregates to real terms. (3) **Combine collinear variables** into a single macro-conditions index if separate effects aren't needed. (4) **Extend the sample** beyond 28 quarters to add independent variation — though Irish macro data will co-trend in most windows. (5) **Do nothing** if the purpose is forecasting: collinearity does not bias predictions, only the attribution to individual coefficients. A complete answer picks a primary recommendation (drop the redundant aggregate and/or difference the data) and defends it, rather than listing options mechanically.

Mark Allocation — Part (c)(ii) (15%)

Band	Criteria
Full credit (13–15)	At least three remedies each with rationale and caveat, targeted at the diagnosed block (names which variable to drop and why, or the specific transformation) AND recognises the do-nothing/forecasting case or the OVB trade-off.
Good (10–12)	Two or three correct remedies with reasoning; targeting or caveats thinner.
Partial (6–9)	Generic remedy list not tied to these variables.
Minimal (0–5)	Single vague suggestion or incorrect remedies.

Question A3 total **15 + 15 + 15 + 15 + 10 + 15 + 15 = 100%**